

# SecurePulse Mobile — Product & Engineering Roadmap



This document outlines the strategic roadmap, milestones, architectural evolutions, and feature iterations for **SecurePulse Mobile** (Enterprise Cybersecurity & SOC Mobile Operations Platform).

## High-Level Vision & Objectives

- **Mission:** Enable cybersecurity analysts, SecOps leads, and CISOs to monitor, triage, investigate, and remediate enterprise security threats anywhere in real-time.
- **Core Pillars:** Zero-Trust Security, Real-Time Low Latency Telemetry, Multi-Model AI Assistance, and Offline-First Resilience.



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## Phase-by-Phase Breakdown

### Phase 1: Core Foundation & Design System (Completed)

- [x] **Material 3 Cyber Theme:** Dark obsidian & neon cyan aesthetic with high accessibility contrast and light mode support.
- [x] **Declarative Routing:** GoRouter shell navigation with 5 primary tabs and nested detail screens.
- [x] **State Management Architecture:** Riverpod 2.x providers with immutable models and reactive stream listeners.
- [x] **Local Encrypted Storage:** Secure token persistence ( `flutter_secure_storage` ) and encrypted Hive database caching.
- [x] **Biometric Integration:** Local hardware-backed authentication (Face ID & Fingerprint via `local_auth` ).

### Phase 2: Live Cloud Backend & Dual Operation Modes (Completed)

- [x] **FastAPI & Render Cloud Integration:** Production API client with JWT auto-injection, request interceptors, and timeout handling.

- [x] **Dynamic Demo Mode:** Seamless offline simulation toggle for instant presentation and testing without cloud dependencies.
  - [x] **Real-Time WebSocket Engine:** Live SIEM/SOC alert streaming with automatic HTTP long-polling fallback.
  - [x] **Automated SAST Triggers:** On-demand repository security scans with vulnerability severities and finding counts.
  - [x] **Executive PDF Generator:** Pure vector PDF report compiler for SOC 2, ISO 27001, PCI-DSS, and HIPAA with cryptographic SHA256 audit stamping.
  - [x] **Integration Test Suite:** 15/15 automated tests verifying API contracts, mode isolation, and WebSocket transformations.
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### **Phase 3: AI Copilot Expansion & Deep Telemetry (Current)**

- [ ] **Interactive Remediation Workflows:** One-tap PR creation for AI-generated vulnerability patches directly to GitHub/GitLab.
  - [ ] **Rich Push Action Handlers:** Interactive push notifications allowing analysts to acknowledge or escalate alerts directly from lock screen.
  - [ ] **Multi-Tenant / Multi-Organization Switching:** Ability to switch between multiple SOC environments and cloud clusters.
  - [ ] **Advanced FICart Visualizations:** Heatmaps of intrusion attempts, geographic attack maps, and MTTR (Mean Time to Remediate) trends.
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### **Phase 4: Enterprise Ecosystem & SOAR Integrations (Next)**

- [ ] **Two-Way SIEM/SOAR Sync:** Direct webhook connectors for Splunk, Elastic SIEM, Wazuh Manager, Microsoft Sentinel, and Jira Security.
  - [ ] **Offline Action Queue:** Queue incident mitigations (IP blocking, firewall rule generation) while offline and automatically dispatch when connectivity restores.
  - [ ] **Custom Rule Builder:** In-app Semgrep and Sigma rule authoring with syntax highlighting and dry-run validation.
  - [ ] **Role-Based Access Control (RBAC):** Fine-grained permissions (CISO, Tier-1 Analyst, Tier-3 Incident Responder, Auditor).
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### **Phase 5: Autonomous Security Mesh (Future)**

- [ ] **Zero-Trust Client Mesh:** Mutual TLS (mTLS) certificate exchange per mobile device for zero-trust API communication.
- [ ] **Voice-Activated Incident Commander:** Natural speech commands ("*SecurePulse, quarantine host prod-db-01 and trigger a high-priority scan*").
- [ ] **Wearable SOC Alerts:** Companion WearOS & Apple Watch apps for critical severity paging and instant acknowledgment.

- [ ] **Decentralized Threat Intelligence:** Peer-to-peer CVE and IOC indicator sharing with MITRE ATT&CK matrix mapping.
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## Technical Debt & Maintenance Schedule

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1. **Dependency Modernization:** Regular quarterly updates for Flutter SDK and pub packages.
  2. **End-to-End Automated Integration:** CI/CD pipeline running Flutter tests on every pull request.
  3. **App Store & Play Store Compliance:** Android 14+ / iOS 17+ permission compliance, privacy manifests, and encrypted keystore rotation.
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